Apply the concepts of pipeling and parallel processing

CO3

1) Benifits of pipeline in a processor

- Increases speed :- Enables instruction faster
- Improves throughput :- Multiple instructions are processed at different stages simultaneously
- Better CPU utilization :- Makes efficient use of processor resources
- Reduces execution time :- Improves overall processor performance.

Increases efficiency :- Allows continuous instruction processing.

2) Difference b/w 3 stages and 5 stage pipeling

-> 3 stages vs 5 stage pipline

Stages :- 3 stage has fetch, decode, execute. 5 stage has fetch, decode, execute, memory, write back

Clock speed :- 3 stages is lowest because the execution stage does too much. 5 stage is higher because each stage is shorter.

Instruction in flight :- 3 has 3, 5 has 5.

Memory Access :- In 3 stage it is done inside execute so load/store takes certain cycles. In 5 stage it has its own memory stage.

Hazards :- 3 stage has fewer and simpler ones. 5 stage has more and needs forwarding and hazard detection.

Hardware and power - 3 Stage is simpler and slower. 5 stage is complex and higher.

3) Explain the relationship of capacitance frequency and power dissipation in single core and multicore.

-> Dynamic Power $P = C \cdot V^2 \cdot f$

C (Capacitance) The total switching capacitance of the chip. More transistors means more C.

f (frequency) -> How often the circuit switches power rises linearly with f

V (Voltage) -> needed to support higher f and since P depends on V^2, power grows roughly as f^3 when voltage is scaled up with frequency.

single core problem :- To get more performance you raise f which needs higher V power and heat shoots up, frequency scaling hits a limit.

Multicore solution :- Instead of one fast core, use several slower cores. Each core runs at a lower f and lower V so power drops significantly.
Total C increases but the savings from lower V and f outweigh it.
Performance is gained through parallelism rather than clock speed.

4) Challenges of system which customer expectations in single and multicore systems.

1. Single core system.
   - slower execution of multiple tasks
   - limited multitasking performance
   - longer response time.
   - Difficulty handling heavier workloads
2. Multi core system
   - Complex task scheduling
   - Higher power consumption
   - Increased system cost
   - Difficulty in balancing workloads across cores

5) What do you mean by hazards explain different kinds of hazards with example.

-) Hazards are problems that occur in a computer processor when instructions cannot execute properly in the expected order.

They may cause delays or incorrect results during instruction execution in a pipeline.

Types of Hazards :-
Mainly there types of Hazards :-

Structural Hazard :-
Occurs when two instructions needs the same hardware resources at the same time
E.g :- Two instructions try to access the same memory simultaneously.

Data Hazard :-
Occurs when the processor forces a branch or jump instruction and does not know which instruction to execute next.

The processor must decide which instruction to execute next.